

Sharayu Rasal

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Education

New York University, Master of Engineering in Computer Science (GPA: 3.67/4.0)

Sep 2024 – May 2026

- **Coursework:** Cloud Computing, Design & Analysis of Algorithms, Software Engineering, Machine Learning, Deep Learning, Human Computer Interaction, Information Privacy & Security, Computer Networks

AISSMS College of Engineering, B.E. in Computer Science (Honors in Data Science)

Aug 2020 – May 2024

- **Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Web Technology, Operating Systems, Natural Language Processing, Database Management Systems, Artificial Intelligence, Blockchain Technology, Business Intelligence

Professional Experience

Research & Development Intern, Bharat Forge Ltd (KCTI)

Pune, India

Jul 2023 – Nov 2023

- Engineered a **real-time Computer Vision pipeline using Python, OpenCV**, and optimized low-level modules for target detection in unmanned systems, achieving an **18% reduction in latency**.
- Implemented and validated **custom data transmission protocols** to improve drone communication reliability under bandwidth constraints.
- Collaborated with **DRDO engineers** to integrate embedded modules, documenting **API specifications and hardware interfaces** for production deployment.

Software Engineering Intern, Posit Source Technologies

Pune, India

Mar 2023 – May 2023

- Designed and automated scalable **ETL data pipelines using Python** to process large-scale logistics datasets, improving data consistency and enabling data-driven decision-making.
- Developed and deployed high-performance **RESTful APIs and interactive dashboards** for real-time analytics, reducing manual reporting workload by **60%**.
- Optimized system performance by refactoring **caching (Redis)** and scheduling logic for a same-day delivery system, **reducing delivery lead time by 30%** and improving task execution efficiency.
- Participated in an **Agile environment**, actively participating in **code reviews, design discussions, and team planning** to align on engineering best practices and ensure **solution scalability**.

Technical Product Intern – Web & Marketing, Oniv Beverages Pvt Ltd

Mumbai, India

Oct 2023 – Feb 2024

- Built and deployed a **customer-facing web application** using React, improving site performance and contributing to a **24% increase in organic reach**.
- Implemented **event instrumentation** via custom JavaScript tracking funnelled into an **AWS Kinesis ingestion pipeline**, enabling real-time analytics and data-driven marketing decisions.

Selected Projects

SmartCache AI (Intelligent Offline Content Recommender) 🔄

- Spearheaded a **full-stack AI content recommendation platform** using **Django REST, React, PostgreSQL, and AWS S3** for personalized offline learning via tag-based curation and multi-source aggregation.
- Engineered a **distributed data processing backend** using **Redis and Celery** for asynchronous background tasks, automated content caching, and multi-source data aggregation.
- Designed an **ML-based recommendation system** using **cosine similarity on vector embeddings** and **top-K tag matching** to suggest personalized content for users.

Multimodal Depression Detection 🔄

- Built a **real-time mental health screening platform** combining **logistic regression, 1D CNN (HRV signals), and facial emotion CNN** for multimodal prediction.
- Engineered **data preprocessing pipelines, APIs, and deployment scripts** for scalable and low-latency inference.
- Achieved **90.7% accuracy (text), 91.6% (HRV), and ROC-AUC 0.95**, demonstrating cross-modal robustness.

LoRA-RoBERTa News Classification App 🔄

- Devised a **parameter-efficient text classifier** fine-tuned with **LoRA-RoBERTa**, achieving **94.6% accuracy** with only **925K trainable parameters (0.7% of total)**.
- Architected a **high-throughput data inference service** using a **FastAPI and Docker stack**, delivering **sub-150 ms predictions** via a **React dashboard** while cutting **GPU memory use by ~95%**.

Skills

Programming: Python, JavaScript, TypeScript, C++, SQL

Backend: Django, FastAPI, Node.js, Express, REST APIs, Flask — **Frontend:** React, HTML5, CSS3, Bootstrap

Cloud & DevOps: AWS (Lambda, Lex, Rekognition, S3, RDS, DynamoDB, OpenSearch, SES, SQS, API Gateway, CloudFormation, Amazon Bedrock EKS), GCP, Docker, Kubernetes (Minikube), CI/CD (GitHub Actions)

Databases: PostgreSQL, MongoDB, Firebase, MySQL — **Tools:** Git, GitHub, Redis, Postman, Docker Hub, VS Code, Swagger, AWS CLI, kubectl

Data Tools: Pandas, NumPy, Matplotlib, Seaborn — **Infrastructure & Monitoring:** Prometheus, CloudWatch, OpenSearch.

Leadership

- **Core Team Member, Google Developers Students Club (GDSC):** Web Dev core team member, organized technical events with 300+ attendees; secured sponsorships from RedBull & GeeksforGeeks.